

Build for Billions

Initial Screening Proposal

Team Name	VoidMain()
Team Leader	Karthik Sudhir Prabhu
Team Members	Krithi P, Mahathi V H, M Sachin Acharya, Pavan M R, Keerthan
College/University	Mangalore Institute of Technology and Engineering (MITE)
Selected Track	Agentic AI For Billions

1. Problem Statement

Millions of people across India — particularly rural citizens, students from low-income families, farmers, and first-generation learners — are entitled to government welfare, education, and financial-inclusion schemes but never access them. The barrier is rarely eligibility; it is friction. Schemes are scattered across dozens of portals in English or complex bureaucratic language, eligibility criteria are unclear, required documents are inconsistently listed, application forms are difficult to navigate for users with low digital literacy, and there is no simple way to track an application once submitted.

Language is a major additional barrier for non-English, non-Hindi speakers (e.g. Kannada, Tamil, Telugu speakers), and voice-first interaction is often the only realistically accessible mode for users who cannot easily read or type complex forms. As a result, a significant share of citizens who qualify for public benefits never claim them — not because the schemes don't exist, but because discovering and navigating them is too hard.

2. Proposed Solution

Sahaay is a voice-first, multilingual AI agent that helps users discover, understand eligibility for, prepare applications for, and track government and education/welfare schemes — with a mandatory human-approval gate before any state-changing action (submitting data or an application on the user's behalf).

Rather than a single chatbot, Sahaay is built as a coordinated multi-agent system:

- **Orchestrator Agent** — the only agent the user directly talks to; holds conversation state and routes intent
- **Discovery Agent** — matches the user's profile (state, occupation, income, stated need) against a curated scheme knowledge base
- **Eligibility Agent** — runs rules-first eligibility checks (income thresholds, occupation match) and calls an LLM only for genuinely ambiguous, free-text criteria — always producing a named reason, never a bare yes/no

- **Document Agent** — checks uploaded documents for completeness and format only (file present, name consistency, expiry) — explicitly never claims to verify authenticity
- **Application Agent** — pre-fills a structured application for human review, never for automatic submission
- **Tracking Agent** — surfaces application status after submission
- **Human-Approval Gate** — a first-class UI step; nothing is ever submitted without explicit user confirmation

Every scheme match the agent surfaces carries a source link, a confidence level, and a last-verified date, so every claim is explainable and checkable rather than a black-box answer.

3. Target Users & Impact

Primary users are rural and low-income citizens, students, farmers, and other underserved populations across India who are entitled to but currently under-utilize government schemes due to language, literacy, and digital-navigation barriers.

The current prototype curates 13 real schemes spanning welfare (MGNREGA, PMAY-G, PMUY, PMMVY), education (NMMS and similar scholarship schemes), and financial inclusion (Stand-Up India, PMJJBY, PMSBY) — chosen because together they serve rural households, women, students, and small entrepreneurs, some of India's largest underserved populations. Because the same architecture generalizes to any scheme with defined eligibility criteria, Sahaay's reach scales with how many schemes and languages are added to its knowledge base, not with how the core system is built.

4. Technology Stack

- **LLM reasoning:** Groq API (Llama 3.3 70B, OpenAI-compatible endpoint)
- **Speech-to-text:** Groq Whisper (large-v3-turbo)
- **Text-to-speech:** Bhashini — Government of India's multilingual language AI initiative
- **Document processing:** Tesseract.js (open-source OCR) for completeness/format checks
- **Agent orchestration:** n8n — self-hosted workflow automation, with one workflow per agent (Orchestrator, Discovery, Eligibility, Documents, Application Prepare/Confirm, Tracking), each triggered via webhook
- **Scheme knowledge base:** a versioned Google Sheet, editable without redeploying code
- **Frontend:** Next.js (App Router) deployed on Vercel, acting as a thin proxy layer with no business logic, calling n8n webhooks server-to-server
- **Development environment:** Google Antigravity, an agentic coding platform used to scaffold the frontend from a structured specification
- **Hosting:** entirely on free tiers (Vercel Hobby, self-hosted n8n via ngrok, Groq's free API tier) — deliberately built to run at zero infrastructure cost

5. Innovation & Scalability

- A first-class trust layer: every claim carries its source, a confidence level, and a freshness date, rather than a single opaque LLM answer
- Rules-first eligibility reasoning: hard criteria are checked with plain logic, and the LLM is invoked only for genuinely ambiguous free-text criteria — reducing hallucination risk in a domain where a wrong eligibility answer has real consequences
- A hard human-approval gate before any submission or data-sharing action — trustworthy by design, not trust asserted after the fact
- A modular, spreadsheet-driven knowledge base that lets new schemes, states, or languages be added without a code deployment
- A genuinely zero-cost architecture, demonstrating the system is viable for student-led or resource-constrained teams to build and operate
- Scalability path: more schemes and states via the knowledge base, more languages via the translation/TTS layer, and WhatsApp/IVR as a longer-term, lower-friction interface

6. Prototype / References

[Team to add: GitHub repository link and/or deployed demo URL before submission]

7. Team Contribution

Karthik Sudhir Prabhu (Team Leader)	<i>[To be filled in]</i>
Krithi P	<i>[To be filled in]</i>
Mahathi V H	<i>[To be filled in]</i>
M Sachin Acharya	<i>[To be filled in]</i>
Pavan M R	<i>[To be filled in]</i>
Keerthan	<i>[To be filled in]</i>